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Research Proposal

Autonomous Environmental Sensor Intelligence: A Multi-Agent MLOps Framework for Real-Time Anomaly Attribution and Drift-Adaptive Monitoring

1. Research Gap

Environmental sensor networks generate continuous, multi-parameter data streams in which anomalies rarely occur in isolation. A single turbidity spike may be sensor noise; simultaneous shifts across turbidity, dissolved oxygen, conductivity, and flow rate at correlated monitoring stations indicate a genuine physical event. Current deployed systems, however, evaluate parameters independently through static rule-based thresholds, producing high false-alert rates and providing no mechanistic account of which sensor parameter caused the anomaly versus which merely responded to it. Three specific knowledge gaps remain unaddressed in the published literature.

Gap 1 — Multi-parameter co-anomaly attribution

Existing multivariate approaches, including the MCN-LSTM architecture of El-Shafeiy et al. [1] and classical outlier detection benchmarks evaluated by Vaquet et al. [2], detect whether a joint anomaly has occurred but do not produce confidence-calibrated attribution across the contributing sensor parameters. Without attribution, field engineers cannot determine which sensor requires inspection, recalibration, or replacement—a critical operational gap in live water quality monitoring networks.

Gap 2 — Drift-adaptive MLOps for environmental IoT

Physical sensors undergo progressive calibration drift under sustained field conditions—a phenomenon well documented in the concept drift literature [3, 4], yet no published MLOps protocol specifies trigger conditions, retraining cadence, or acceptable performance recovery windows specifically for multi-parameter environmental sensor systems operating on live field data. Existing drift mitigation work [3, 4] addresses synthetic benchmarks and federated IoT settings; validated protocols for real environmental monitoring deployments do not exist.

Gap 3 — LLM-generated incident summaries for non-specialist responders

When an automated system detects a multi-parameter anomaly, the information format delivered to a field engineer or regulatory stakeholder directly determines response latency and decision accuracy. Liu et al. [5] and Nie & Liu [6] identify natural language generation as a high-potential but empirically unvalidated pathway for environmental monitoring communication; no published study has measured whether retrieval-augmented LLM summaries grounded in historical sensor baselines and regulatory thresholds produce faster or more accurate intervention decisions than conventional threshold-alert formats.

This proposal addresses all three gaps using 70,000+ real sensor records from eight active ASU SSEBE monitoring stations across 14 environmental parameters, a production dataset with labeled operational context unavailable to published studies relying on synthetic or single-parameter benchmarks.

2. Research Questions

RQ1 — Attribution accuracy: Can a LangGraph-orchestrated multi-agent system, combining LSTM temporal modeling with Isolation Forest ensemble scoring across 14 co-varying parameters, achieve statistically significant improvement in anomaly attribution precision and recall over single-model statistical baselines on real environmental sensor data—and does performance hold under documented calibration drift conditions? Falsifiable claim: the multi-agent system achieves $F1 > 0.80$ on a held-out test set where the rule-based baseline achieves $F1 < 0.65$ on the same evaluation harness.

RQ2 — Drift-adaptive retraining: What sensor-specific and environmental-context features most reliably predict model performance degradation under calibration drift in a multi-parameter IoT system, and can an Evidently AI-monitored, Airflow-orchestrated automated retraining pipeline recover baseline F1 within a defined latency window following a detected drift event? Falsifiable claim: the drift-adaptive pipeline recovers to within 5% of pre-drift F1 within 48 hours of drift detection on the live sensor dataset.

RQ3 — Stakeholder response quality: Do natural-language incident summaries generated by a retrieval-augmented LLM agent—grounding outputs in historical sensor baselines and EPA regulatory thresholds—produce measurably faster or higher-accuracy intervention decisions than threshold-alert formats in structured expert evaluation? Falsifiable claim: expert evaluators reach correct intervention decisions at least 25% faster when presented with LLM-generated summaries versus numeric alert tables on a standardized 20-case evaluation set.

3. Existing Infrastructure

This is not a proposal beginning from scratch. The following production systems are operational within the SSEBE lab as of May 2026:

System	Status and evidence
IoT data pipelines	Validated Python and SQL ingestion for 70K+ records across 14 sensor parameters stored in PostgreSQL with model-ready schemas; 40% improvement in false-alert reduction over rule-based baselines
Engineering API	PostgreSQL-backed access APIs with date-range filtering and CSV/JSON export; 35% reduction in retrieval latency for downstream research teams
Analytics dashboards	Tableau dashboards tracking environmental KPIs across 8 stations; 3× reduction in insight turnaround for project leadership
Geospatial ELT pipelines	Integration of CWNS 2022 and NHDPlus federal datasets across 10,000+ records; 60% processing time reduction via automated schema reconciliation and validation (prior SSEBE role, Feb–Oct 2025)

4. Methodology — 8-Week Build Plan

Each phase produces a testable intermediate deliverable to ensure the research trajectory remains falsifiable and on schedule.

Phase	Weeks	Deliverable	Addresses
1	1–2	Kafka ingestion layer simulating real-time sensor events; Spark Structured Streaming for windowed feature aggregation over 14 parameters	RQ1, RQ2 foundation
2	3–4	LSTM temporal anomaly model + Isolation Forest ensemble; MLflow model registry with champion/challenger versioning; held-out test set evaluation harness	RQ1 primary
3	4–5	LangGraph multi-agent system: Sensor Drift Agent, Threshold Breach Agent, Incident Report Agent (RAG-grounded LLM summaries), Stakeholder Escalation Agent	RQ1, RQ3 primary
4	6–7	Evidently AI drift monitoring; Airflow DAG for automated retraining on calibration drift signals; GitHub Actions CI/CD with evaluation gates; RQ2 validation experiment	RQ2 primary
5	8	FastAPI serving endpoint; Docker + Kubernetes deployment; Grafana + Prometheus observability; structured expert evaluation study for RQ3	RQ3, deployment

5. Expected Contributions

Contribution 1 — A publicly released benchmark dataset. The first multi-parameter environmental IoT anomaly dataset with ground-truth labels across 14 sensor parameters, 8 monitoring stations, and documented calibration drift events. Enables reproducibility and direct comparison for future work. Intended for release on Zenodo concurrent with publication.

Contribution 2 — A confidence-calibrated anomaly attribution framework. A validated multi-agent approach that distinguishes causal from symptomatic sensor parameters in co-anomaly events, evaluated against rule-based and single-model baselines on the benchmark above. Falsifiable target: $F1 > 0.80$ on held-out test set.

Contribution 3 — A drift-adaptive MLOps protocol. A validated retraining protocol—Airflow-orchestrated, Evidently-monitored—specifying trigger conditions, retraining frequency, and performance recovery time under documented sensor calibration drift patterns. Directly transferable to other environmental IoT deployments.

Contribution 4 — An open-source deployment framework. A fully documented, containerized system (Docker + Kubernetes) released under MIT license, enabling replication at other smart water utility or climate monitoring installations. Intended as a reusable foundation for future NSF or EPA-funded research proposals.

6. Publication Target

Primary venue	Environmental Modelling & Software (Elsevier, IF 4.9) — the field’s primary outlet for computational methods in environmental science, with an active ML and data-driven approaches track
Working title	Confidence-Calibrated Anomaly Attribution in Multi-Parameter Environmental Sensor Networks: A Multi-Agent MLOps Framework with Drift-Adaptive Retraining
Preprint	arXiv (cs.LG + eess.SP) on system completion — enabling immediate visibility and citation before formal peer review concludes
Secondary use	Live demo system deployable for NSF or EPA grant reviewers as evidence of active, production-grade research infrastructure within the SSEBE lab

7. Candidate Qualifications

Dimension	Evidence
Domain depth	IIT Kharagpur B.Tech Civil Engineering + M.Tech Environmental Engineering (Dual Degree, GPA 3.8); active SSEBE research role since October 2025 — a combination no ML-only candidate can replicate
Existing infrastructure	70K+ real sensor records already in PostgreSQL; ingestion pipelines, anomaly detection, and BI layer operational — zero data acquisition delay
Production ML experience	Anomaly detection, ELT pipelines, and API layers at SSEBE; classification and churn models at AWL Metaverse (500K+ records); marketing analytics at Dharte Inc.
Full-stack ML capability	Python, SQL, Spark, MLflow, FastAPI, Docker, AWS, GCP, TensorFlow, PyTorch, HuggingFace — all in active use across current and prior research roles
Agentic AI architecture	Shipped multi-agent systems: Mycelium AI (multi-agent knowledge helpdesk with Fetch.ai uAgents), Aria VisionAI (LangGraph voice assistant), TrustMed-AI (medical RAG with 99% uptime)
Availability	Full-time research focus available for Summer 2026 (May–August); no coursework conflicts during this period
Work authorization	F-1 student; eligible for Curricular Practical Training (CPT) for summer RA appointment

8. Request

I am requesting a Research Assistant appointment of 20 - 40 hours per week for Summer 2026 (May–August) to pursue this work formally under faculty supervision. The deliverables a production-grade autonomous monitoring platform, a benchmark dataset, and a manuscript submission to Environmental Modelling & Software—are defined, scoped to the available timeline, and directly extend infrastructure already in place within the lab. The open-source framework and demo system are additionally available as supporting assets for the next SSEBE grant review cycle.

Full project documentation, GitHub repositories, and working code are available upon request. I would welcome the opportunity to discuss whether this direction aligns with the lab’s current research priorities.

References

- [1] El-Shafeiy, E., Alsabaan, M., Ibrahim, M. I., & Elwahsh, H. (2023). Real-Time Anomaly Detection for Water Quality Sensor Monitoring Based on Multivariate Deep Learning Technique. *Sensors*, 23(20), 8613. <https://doi.org/10.3390/s23208613>
- [2] Vaquet, V., Hinder, F., & Brinkrolf, J. (2024). Towards Reliability in Smart Water Sensing Technology: Evaluating Classical Machine Learning Models for Outlier Detection. *Sensors*, 24(13). <https://doi.org/10.3390/s24134333>
- [3] Hinder, F., Vaquet, V., & Hammer, B. (2024). One or two things we know about concept drift — a survey on monitoring in evolving environments. Part A: detecting concept drift. *Frontiers in Artificial Intelligence*, 7. <https://doi.org/10.3389/frai.2024.1330257>
- [4] Mavromatis, I., Carnelli, P., Khan, A., Bhatt, U., & Piechocki, R. (2024). FLAME: Adaptive and Reactive Concept Drift Mitigation for Federated Learning Deployments. *arXiv:2410.01386*. <https://arxiv.org/abs/2410.01386>
- [5] Liu, Q., et al. (2025). Large language models: Tools for new environmental decision-making. *Environmental Research*, 267, 120632. [https://doi.org/10.1016/S0301-4797\(25\)00349-4](https://doi.org/10.1016/S0301-4797(25)00349-4)
- [6] Nie, Q., & Liu, T. (2024). Large language models for environmental modeling: Framework, capabilities, constraints. *Environmental Science & Technology*. <https://doi.org/10.1021/acs.est.3c01106>